

CSE221: Algorithms

Lab-01

Fall 2025

Action required for Students

[Google Form] CSE221 Lab Policy:

- The lab policy file can be found here - [Link](#)
- Please complete the lab policy acknowledgement form - [Link](#)

[Google Form] Codeforces Account Information: (Action required)

- Open an account at <https://codeforces.com/>
- Please submit your codeforces details here - [Link](#)

Necessary Resources [Bookmark the Link]

Student Drive Folder: [Link](#)

The folder contains:

- Lab Policy
- Sample Quiz Checkers
- Environment Setup for Quiz Checker
- Previous Semesters' Lab Quiz Questions

Assignment-01 Links:

codeforces Invitation Link:

<https://codeforces.com/contestInvitation/0933ec323333af14cd21eb6689732753f301afff>

Alternative Link: <https://codeforces.com/contests/643746>

[The alternative link will **only work** after you have successfully registered using the first link.]

Deadline: Oct 26, 2025 2:00 PM

No submission can be considered after the deadline ends.

Learning Outcome of this Lab:

- Understand the basics of Online Judge platforms.
- Become familiar with standard Input/Output (I/O) formats used in Online judge
- Learn about common judge verdicts, including Accepted (AC), Wrong Answer (WA), Time Limit Exceeded (TLE), and Runtime Error (RE).
 - For a **1-second** time limit, most OJs allow approximately **10^8 operations**.
 - For python select PyPy over Python if available, as PyPy is faster. However, PyPy consumes more memory than Python.
- Review $O(n^2)$ sorting algorithms which have already been covered in the previous courses such as Bubble Sort, Selection Sort, and Insertion Sort.
 - Learn to sort with multiple parameter
 - Understand the use case of stable sort
- Explore Fast I/O techniques (**self-study**):
 - **Python:** Using `sys.stdin` for faster input handling.
 - **C++:** Using `ios_base::sync_with_stdio(false);` and `cin.tie(nullptr)`
 - **Java:** Using `BufferedReader` for efficient input reading.
 - The fast i/o can reduce program execution time on Online Judge platforms.
 - **Fast I/O Example:** [Python](#), [C++](#), [Java](#) [click Expand to view the code snippet]
 - **Slow I/O vs Fast I/O Sample:** [Python and Java](#)